

Lithium ion conductive properties of aliphatic polycarbonate

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ABSTRACT

Properties of polymer electrolytes composed of poly(ethylene carbonate) [PEC] and lithium bis(trifluoromethane sulfonimide) [LiTFSI, $\text{LiN}(\text{CF}_3\text{SO}_2)_2$] have been analyzed using Raman spectroscopy, dynamic mechanical analysis (DMA) measurements, and ac impedance spectroscopy. The lithium ion transference number of a PEC based polymer electrolyte is 0.4, which is 4 times higher than that of a poly(ethylene oxide)[PEO] based polymer electrolyte. The glass transition temperature of a PEC based polymer electrolyte decreases with the increase of LiTFSI concentration above 0.2 molar ratio of LiTFSI to repeat unit of the carbonate in PEC. Raman spectra show that the abundance ratio of undissociated LiTFSI increases at LiTFSI concentration above 0.2 molar ratio. Moreover, ionic conductivity of a PEC based polymer electrolyte increases with the increase of LiTFSI concentration above 0.2 molar ratio. The highest ionic conductivity is 0.47 mS cm^{-1} at 20°C .

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1. Introduction

A solid polymer electrolyte is an important material for all-solid electrochemical devices, such as lithium ion batteries, electrochromic displays, and chemical sensors. In particular, all-solid lithium ion batteries are attractive applications for a solid polymer electrolyte. Since Wright et al. [1] discovered the complex formation of polyethylene oxide (PEO) and electrolytic salt, and Armand et al. [2] discovered the possibility of applying that complex as a battery electrolyte, polymer electrolytes have been the subject of much research.

Since the biggest disadvantage of a polymer electrolyte is its low ionic conductivity, elucidating the mechanism by which ionic conductivity is expressed and increasing ionic conductivity are being investigated as key issues. The conductive mechanism of the ions in a polymer electrolyte was studied in detail by Armand et al. [3]. They investigated the influence of the proportions of crystal phase and amorphous phase in PEO/ LiCF_3SO_3 on ionic conductivity, and they subsequently found that increasing the amorphous-phase proportion improves ionic conductivity.

As methods for increasing ionic conductivity, two approaches have mainly been investigated, namely, (i) suppressing crystal generation in the polymer by adding filler and (ii) creating a comb-shaped polymer framework, and high ionic conductivity has thus been achieved by taking these approaches. As for the first approach, a method of suppressing crystallization of the polymer by adding a fine ceramic powder (with micrometer-order particle diameter) to a PEO based polymer electrolyte has been reported [4,5]. Moreover, by mixing

nanometer-order Al_2O_3 and SiO_2 as a ceramic powder to a concentration of 10 wt.% in PEO containing an electrolytic salt, Scrosati et al. successfully increased ionic conductivity several times in the temperature range of 60 to 80°C [6]. The mechanism of ionic conductivity is continuing to be analyzed [7]. Moreover, as for the second approach, Watanabe et al. combined a PEO based polymer composed of ethylene-oxide chains to main chains and side chains and achieved ionic conductivity of $10^{-4} \text{ S cm}^{-1}$ at room temperature [8–10]. The relaxation time of lithium ions coordinated with the side chains of the PEO is shorter than that of the main chains of the PEO, and the magnitude of the mobility of the side chains is thought to be the reason for this difference in relaxation times [11]. Lately, Wang et al. have reported an enhanced ionic conductivity in a PEO based polymer electrolyte by controlling the ionic conductive pathway via a core-shell structure [12]. Saikia et al. have reported organic–inorganic hybrid polymer electrolytes based on polyether diamine, alkoxysilane, and trichlorotriazine, which have exhibited ionic conductivity up to $1.6 \times 10^{-4} \text{ S cm}^{-1}$ at 30°C [13].

Nevertheless, according to investigations on improving ionic conductivity by lowering glass-transition temperature in the same manner as these approaches mentioned above, ionic conductivity has reached the level of $3 \times 10^{-4} \text{ S cm}^{-1}$ (room temperature) [14]; however, the practical level ($10^{-3} \text{ S cm}^{-1}$ at room temperature) [15] has not been attained yet.

Furthermore, in addition to achieving high ionic conductivity, improvement of lithium ion transference number of polymer electrolyte is being researched. Ion conduction within a polymer electrolyte is generally made up of anions and cations. In regard to lithium ion batteries, lithium ions diffuse between the anode and cathode of the battery, so increasing the conductance of the lithium ions is important. Lithium ion conductance is the product of ionic conductivity of the polymer

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electrolyte and lithium ion transference number. Consequently, to improve lithium ion conductance, it is necessary to increase ionic conductivity and lithium ion transference number concurrently. To improve lithium ion transference number, research aimed at improving lithium ion transference number by means of anion traps, for example, introducing boron, which has electron attractivity due to its empty 2p orbital, into the polymer framework, is ongoing [16,17]. For example, in the case that LiCF_3SO_3 is used as the electrolytic salt, although lithium ion transference number is increased from 0.10 to 0.75 by dosing with anion traps, ionic conductivity is below $7.2 \times 10^{-6} \text{ S cm}^{-1}$ at room temperature, and improving ionic conductivity is a major challenge. To practically apply a polymer electrolyte in the manner described above, it is necessary to not only improve ionic conductivity but also increase lithium ion transference number; in other words, it is required to improve effective lithium ionic conductivity, namely, the product of ionic conductivity and lithium ion transference number.

Aiming at this improvement of effective lithium ionic conductivity, we have focused on a donor-type polymer framework. As previously described, lithium ions link in a coordinated manner via the oxygen atoms in the polymer framework, and ions are transported by thermal vibration of the polymer framework. We consider that, in the case of a PEO based polymer electrolyte, coupling of ether oxygen in the polymer framework and lithium ions is strong, so ion motion is difficult, and lithium ion transference number is low. Given that fact, we targeted improving ionic conductivity and lithium ion transference number by introducing a carbonate, which forms coordination bonds under a weaker force, into the polymer framework. As a polymer containing carbonate, polyethylene carbonate (PEC), one type of aliphatic polycarbonate, was used. As a low-donor-concentration functional group that can reduce coordinate bonding of polymer chains and lithium ions, a carbonate base was introduced in an attempt to increase ionic conductivity.

In this study, to confirm an introduction of a carbonate functional group into a polymer framework, we investigated lithium ion transference numbers and ionic conductivities of PEC based polymer electrolytes. Furthermore, to discuss a solubility and coordination model of the polymer electrolytes, we evaluated mechanical properties and Raman spectroscopy.

2. Experimental

Poly(ethylene carbonate)[PEC] (Mw 50,000, PAC Polymers Inc.), poly(ethylene oxide)[PEO] (Mw 100,000, Aldrich Inc.) and lithium bis(trifluoromethane sulfonimide) [LiTFSI, $\text{LiN}(\text{CF}_3\text{SO}_2)_2$] (Aldrich, Inc.) were dried at 80 °C, under reduced pressure for over 24 h before use. Dimethylcarbonate [DMC], diethyl carbonate [DEC] and diethyl ether [DEE] (Wako Pure Chemical Industries, Ltd.) were used as received conditions. The molecular unit of $[\text{CH}_2\text{CH}_2\text{O}(\text{C}=\text{O})\text{O}]$ in the PEC and $[\text{CH}_2\text{CH}_2\text{O}]$ in the PEO was defined as the repeat unit so as to determine the molar ratios of LiTFSI to the polymer. In this paper we define the “molar ratio” as follows:

Molar ratio: LiTFSI / molecular unit of $[\text{CH}_2\text{CH}_2\text{O}(\text{C}=\text{O})\text{O}]$ in PEC and LiTFSI / molecular unit of $[\text{CH}_2\text{CH}_2\text{O}]$ in PEO

For preparation of the LiTFSI solution consisting of LiTFSI and DEC or DEE, LiTFSI was dissolved in DEC or DEE at the molar ratio of 0.4 LiTFSI per functional group of the solvents, a carbonate in DEC or an ether in DEE. Also, for preparation of the LiTFSI/PEC electrolyte and LiTFSI/PEO, PEC, PEO and LiTFSI were dissolved in DMC separately. The PEC and PEO solution was kept at 40 °C for 24 h. The solutions were mixed at the required molar ratio of lithium salt to repeat units of the PEC and PEO. After continuous stirring at room temperature, the solution was cast on a PTFE sheet. The solution was allowed to stand at room temperature for a further 24 h to facilitate degassing. The gelatinous polymer solution on the PTFE sheet was dried under reduced pressure at 80 °C

for over 24 h. After drying, the samples were rapidly cooled from 80 °C to 20 °C. An obtained film thickness was around 0.1 mm.

Lithium ion transference number measurements were performed according to the procedure developed by Bruce et al. [18]. In this work, we followed a previous report [19] using a frequency analyzer (Solartron 1260) and a potentiostat analyzer (Solartron 1287). The evaluation method is combined dc and ac analysis methods on the symmetrical cell with nonblocking lithium electrodes in a glove box filled with Ar gas. The dc measurements of initial (I_0) and the steady-state (I_{ss}) currents were made by polarizing the cell at the voltage bias of 10 mV. The ac impedance measurements were carried out before and after applying a bias, to evaluate the value of the interfacial resistance at the beginning and at the end of the measurement; the lithium ion transference number is given by the equation, namely,

$$t_{\text{Li}^+} = \frac{I_{ss}(\Delta V - I_0 R_0)}{I_0(\Delta V - I_{ss} R_{ss})}, \quad (1-3)$$

where t_{Li^+} is lithium ion transference number, I_{ss} is steady-state current, ΔV is applied voltage, I_0 is initial current, R_0 is initial interfacial resistance, and R_{ss} is interfacial resistance at the end of the dc measurement.

The ^7Li nuclear magnetic resonance (NMR) chemical shifts in the DEC, DEE, or a 1:1 molar ratio mixture of both solvent solutions at the molar ratio of 0.4 LiTFSI per functional group of the solvents were measured using a JEOL JNM-GSX400 Fourier transform NMR spectrometer (155.37 Hz for ^7Li nucleus) with an external standard of 1 mol dm^{-3} LiCl deuteration solvent.

Linear sweep voltammetry was used to determine an electrochemical stability window of a LiTFSI/PEC electrolyte. The measurement was carried out using a three-electrode electrochemical cell consisting of a stainless working electrode and a lithium and a counter electrode in a glove box filled with Ar gas. A potentiostat analyzer (Solartron 1287) was utilized under the scan rate of 10 mV s^{-1} and the potential range of 3–10 V vs. Li/Li^+ .

The Raman scattering spectra were measured with a RFT-400 Fourier transform Raman spectrometer (JASCO Corporation) using a 1.024 μm excitation beam. The power from the laser was 800 mW. The scattered light was collected in a 180° configuration. All spectra were accumulated for 100 scans with a resolution of 1 cm^{-1} .

Dynamic mechanical analysis (DMA) measurements of the polymer electrolyte were performed using a DVA-200 dynamic mechanical analyzer (IT Keisoku Seigyō Co.) in shear mode at 10 Hz. The T_g values of the samples were estimated by taking account of the peak in temperature dependence of $\tan\delta$.

Ionic conductivity measurements were carried out using a frequency analyzer (Solartron 1260) and a potentiostat analyzer (Solartron 1287) in a glove box filled with Ar gas. The samples were sandwiched between two Li foil electrodes. The cell constants for the thickness of the polymer electrolyte and the area of the electrode were determined by micrometrical measurements for the electrolyte–electrode contact surface and the distance between the electrodes. Two sheets of a polyimide separator (UBE Industries, Ltd.) were also installed between the two Li electrodes, and the polymer electrolyte was kept inside the space surrounded by the separators and the two Li electrodes. Therefore, the thickness of the polymer electrolyte of the cell was fixed by that of the sheet of the polyimide separator. Two-electrode ac impedance spectroscopy, i.e., using working and counter electrodes, was employed to measure ionic conductivities under the measurement conditions of 1 Hz to 1 MHz ac ranges of 10 mV amplitude. The ac signals were imposed at the open circuit voltage. The samples were heated from 20 to 80 °C and kept at 80 °C for 30 min. They were then cooled to –20 °C. The specific ionic conductivity for LiTFSI/PEC electrolytes was calculated from the equation,

$$\sigma = \frac{t}{AR} \quad (1)$$

where t is the thickness of the electrolyte film, A the film area, and R the film resistance.

3. Results and discussion

Fig. 1(a) shows the results of potential-step chronoamperometry of the PEC based polymer electrolyte or the PEO based polymer electrolyte at the molar ratio of 0.4. According to the previous report [19], this evaluation method sandwiches an electrolyte between lithium metal and applies a dc voltage of 10 mV to it. It is clear that the polymer electrolyte with the PEC system has high steady-state current. Fig. 1(b) shows the impedance profile of the two samples before and after the potentiostatic tests. The lithium ion transference number was obtained from Eqs. (1–3). The transference number in the PEO or PEC case is 0.1 or 0.4, respectively. It is clear from the table that the PEC based polymer electrolyte has the higher lithium ion transference number, namely, 0.4 for the PEC system compared to 0.1 for the PEO system.

To investigate the difference between lithium ion transference numbers in the case of the PEO and PEC based polymer electrolytes, in regard to ether oxygen and the carbonate, which are possibly lithium ion coordination sites in each polymer, their donor characteristics were evaluated. The charge distribution in a solvent molecule is changed by the solvation of a salt [20]. In general, this change induces a chemical shift of the NMR spectra in terms of the solvent molecule's ability to donate electrons. This ability is considered to be the parameters of Lewis basicity, donor numbers, and dielectric constants. ^7Li NMR measurement was performed, and from the viewpoint of a medium containing ether oxygen and/or a carbonate base, the difference in electron states of ^7Li in each solvent namely, diethyl ether, diethyl carbonate, or a 1:1 molar ratio mixture of the two, was evaluated. It is clear from Table 1 that a “chemical shift” occurs in each solvent. The lithium-salt concentration is 0.4 mol per 1 mol of repeating unit of the solvents. The degree of chemical shift is ranked in the following order: diethyl carbonate, 1:1

Table 1

Peaks of ^7Li of LiTFSI in ether, carbonate, or mixture solvents.

Composition of electrolyte			Chemical shift of Li-ion (ppm)
Diethyl carbonate	Diethyl ether	LiTFSI (molar ratio per solvent)	
100	0	0.4	−0.411
50	50	0.4	−0.424
0	100	0.4	−0.454

mixture of both solvent, diethyl ether, and the shift in the ether solvent is confirmed by the peak on high-magnetic-field side. This means that ^7Li is shielded more in the ether solvent than in the carbonate solvent, and the electron density is high. As a result, it is conceivable that the ether oxygen has a stronger donor characteristic than the carbonate base, and a carbonate polymer with a carbonate base has a weaker donor characteristic than that of a polymer with ether oxygen. For that reason, electrostatic interaction between lithium ions and the polymer is weaker, and thus movement of lithium ions is easier, in the PEC based polymer electrolyte than that in the PEO based polymer electrolyte. Consequently, PEC based polymer electrolyte which has a weaker donor characteristic than that of a polymer with ether oxygen shows a higher Li^+ transference number.

The results of dynamic mechanical analysis of the PEC based polymer electrolyte are plotted in Fig. 2. The temperature dispersions of storage modulus, loss modulus, and loss tangent ($\tan\delta$) are evaluated in this figure. Hereafter, the temperature expressing the $\tan\delta$ peak is taken as the glass-transition temperature, T_g .

Fig. 3 shows the LiTFSI concentration dependence of T_g and the storage elastic modulus at 20 °C for LiTFSI/PEC electrolytes. Figs. 4 and 5, respectively, show the storage elastic modulus and the loss elastic modulus as functions of temperature for the polymer electrolytes. In Fig. 3, both the T_g 's and the storage elastic modulus for the polymer electrolytes increase with the addition of LiTFSI up to 0.2 molar ratio per repeat unit and the values decrease with further increase of the LiTFSI concentration. Angell and Forsyth et al. demonstrated that poly ether based polymer electrolytes such as a poly(propylene oxide) or poly(ether urethane) also showed a T_g increase with the addition of LiCF_3SO_3 and T_g showed a maximum value [21,22]. They ascribed the trend of these T_g increases versus LiCF_3SO_3 concentration to an ionic cross-link. The coordination number for lithium in PEO was 4 or 5 taking into account spectroscopic analyses [23]. On the other hand, for the carbonate solvent, reported by Capiglia et al., the coordination number for lithium was 4 using Raman spectroscopy [24]. Therefore, the coordination number in carbonate solvent was equivalent to 0.25 molar ratio per repeat unit. These results showed that the stoichiometric molar ratio of

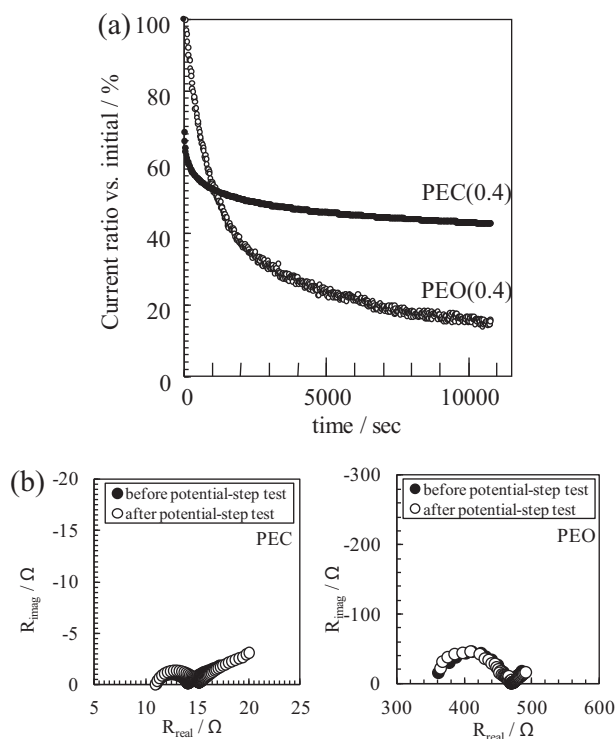


Fig. 1. Potential-step chronoamperometry (a) and impedance spectra variation before after the potential-step tests (b) using a PEC-based polymer electrolyte and a PEO-based polymer electrolyte. The LiTFSI concentration is 0.4 per repeat unit of PEC or PEO.

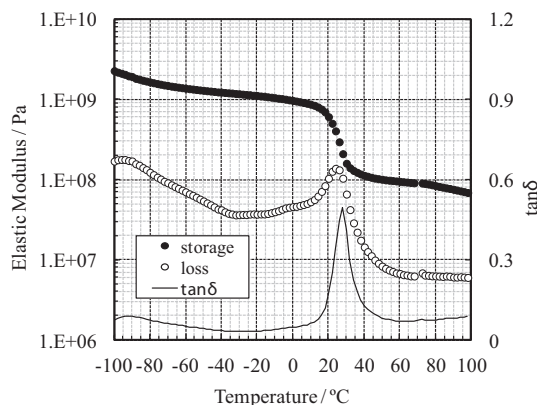


Fig. 2. Dynamic mechanical thermal analysis measurements of a PEC-based polymer electrolyte.

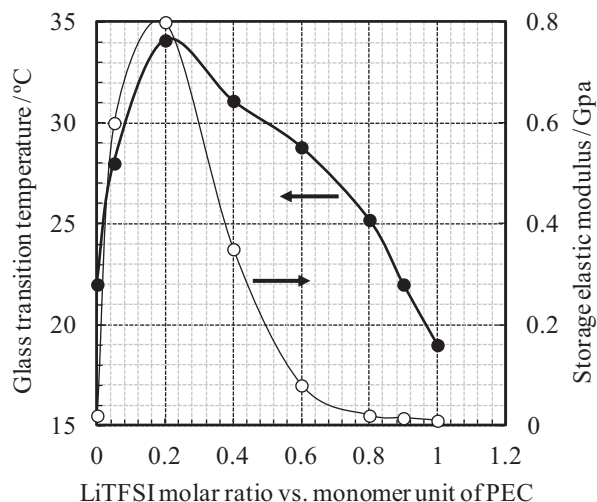


Fig. 3. Salt concentration dependence of the glass-transition temperature or the storage elastic modulus for LiTFSI/PEC electrolytes.

lithium salt to carbonate repeat unit in the polymer complex was around 0.2–0.25. Therefore, it is considered that, at concentrations up to 0.2 molar ratio per repeat unit of PEC, LiTFSI forms a complex with PEC and that at concentrations over 0.2 molar ratio per repeat unit of PEC, the mechanism of the relationship between T_g and LiTFSI concentration is not clear for LiTFSI/PEC polymer electrolyte.

In order to analyze the relationship between T_g and LiTFSI concentration in Fig. 3, Raman spectra of the polymer electrolytes, demonstrated in Fig. 6, are examined in detail. As shown in Fig. 6, the values around 750, 920, 1466, and 1763 cm^{-1} in particular, varied with the addition of LiTFSI. These values, respectively, correspond to the bond vibration modes for the symmetric deformation band of the CF_3 of the LiTFSI, the mixed modes for the symmetric C–O–C stretching and the CH_2 scissoring of the PEC, the CH_2 scissoring mode of the PEC, and the C=O stretching mode of the PEC. The significant change in the observed Raman spectra at 750 cm^{-1} which is assigned to the symmetric deformation band of the CF_3 ($\delta_s(\text{CF}_3)$) of the anion of LiTFSI, $\text{N}(\text{SO}_2\text{CF}_3)_2^-$ is notable. Fig. 7 shows the magnified Raman spectra in the regions of 700–800 cm^{-1} . LiTFSI concentration dependence of the $\delta_s(\text{CF}_3)$ peak for LiTFSI/PEC electrolytes is shown in Fig. 8. The upper horizontal axis in Fig. 8 shows a molar ratio of repeating unit of PEC per LiTFSI. It is

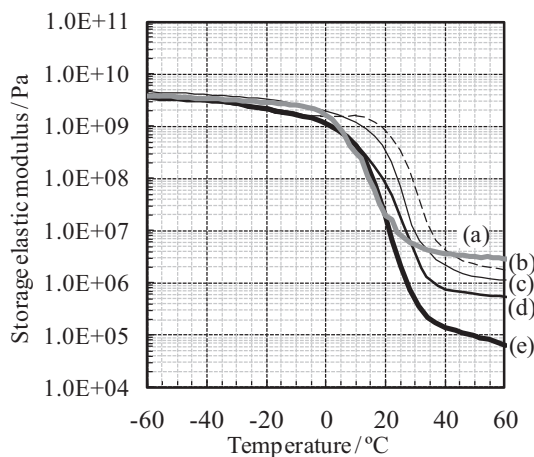


Fig. 4. Salt concentration dependence of storage elastic modulus spectra for LiTFSI/PEC electrolytes. For (a), (b), (c), (d), and (e), LiTFSI concentration per repeat unit of PEC is 0, 0.2, 0.4, 0.6, and 0.8, respectively.

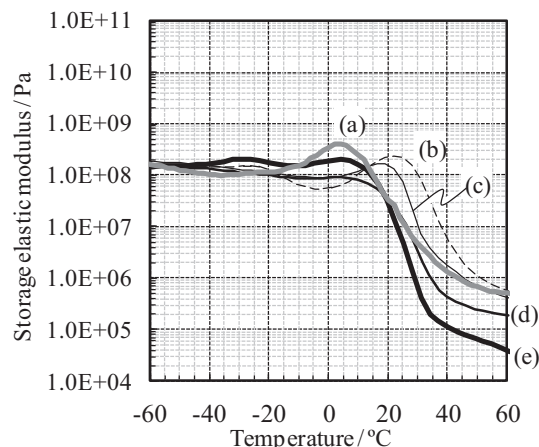


Fig. 5. Salt concentration dependence of loss elastic modulus spectra for LiTFSI/PEC electrolytes. For (a), (b), (c), (d), and (e), LiTFSI concentration per repeat unit of PEC is 0, 0.2, 0.4, 0.6, and 0.8, respectively.

known that this vibration mode is influenced by the dissociation state of LiTFSI and that the peak at 746 cm^{-1} is attributed to the $\delta_s(\text{CF}_3)$ in crystalline LiTFSI [25,26]. Capiglia et al. reported that the calculated wavenumber of $\delta_s(\text{CF}_3)$ for uncoordinated $\text{N}(\text{SO}_2\text{CF}_3)_2^-$ was 737 cm^{-1} , which was lower than that of the LiTFSI crystal state [24]. In the regions above 0.4 molar ratio in Fig. 8, these wavenumber shifts from LiTFSI only have not been reported previously. Taking account of all the considerations described above, the wavenumber shifts appearing in the region seem to be due to an ionic aggregation of the LiTFSI. That is, it is considered that the LiTFSI dissolved in the PEC up to the molar ratio of 0.2 makes cross-link by a coordination between the lithium ion and the carbonate groups in PEC. Thus, as shown in Fig. 3, both the T_g 's and storage elastic moduli for the samples exhibit maximum values at the LiTFSI concentration of 0.2 mol per repeat unit. Meanwhile, in the case of the samples over 0.2 mol per repeat unit, both the T_g 's and storage elastic moduli decrease, because of the presence of an excess amount of a non-dissociative LiTFSI acting as a plasticizer. The plasticizing effect of LiTFSI is also confirmed by other study [27]. In Fig. 6, when the LiTFSI

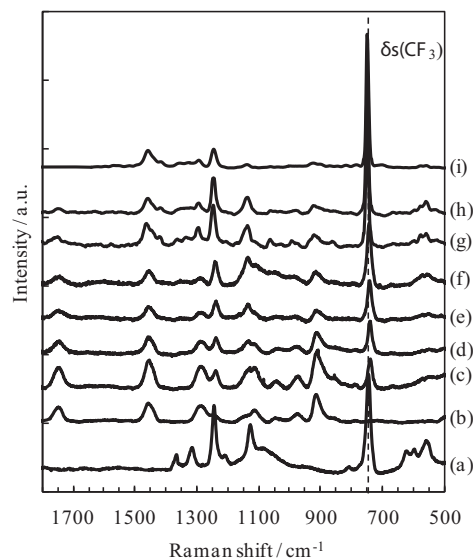


Fig. 6. Raman spectra for LiTFSI/PEC electrolytes, LiTFSI and PEC. For (a) LiTFSI, (b) PEC, (c)–(i) LiTFSI/PEC electrolytes ((c):0.05, (d):0.1, (e):0.2, (f):0.4, (g):0.6, (h):0.8, (i):1.0 per repeat unit of PEC).

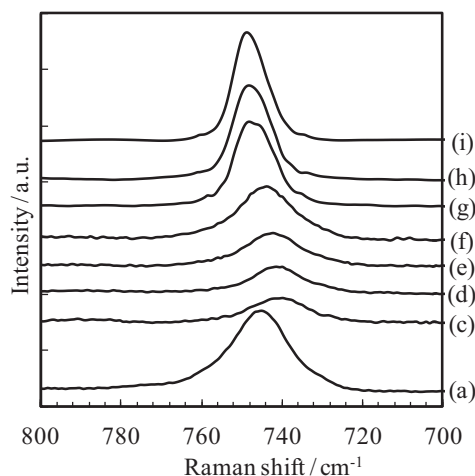


Fig. 7. Raman spectra for LiTFSI/PEC electrolytes. For (a) LiTFSI, (c)–(i) LiTFSI/PEC electrolytes ((c):0.05, (d):0.1, (e):0.2, (f):0.4, (g):0.6, (h):0.8, (i):1.0 mol per repeat unit of PEC).

concentration is over 0.6 mol per repeat unit, a weak peak at 844 cm^{-1} can be perceptible to be observed. At the same time, a weak peak at 1409 cm^{-1} is also observed over the molar ratio of 0.6. The peak intensity at 1763 cm^{-1} decreases with the increment of the molar ratio. It is well known that an increase in charge polarization around the C=O induces a reduction of the intensity of the spectra [28]. Therefore, in this work, as shown in Fig. 6, the observed peak intensity changes at 1763 cm^{-1} is also affected by the charge polarization around the C=O. These findings and the DMA results reveal that there is a dissociation of the LiTFSI in PEC, even though the electron-donating ability of a carbonate functional group in PEC is weaker than that of an ether functional group in PEO.

The temperature dependence of ion conductivity of PEC based polymer electrolyte is plotted as an Arrhenius plot in Fig. 9. As LiTFSI concentration increases, ion conductivity increases, and when it is 0.8 mol, ion conductivity reaches 0.47 mS/cm at $20\text{ }^{\circ}\text{C}$. Moreover, the curve of ion-conductivity temperature dependence shows behavior called “upward-convex Vogel–Tammann–Fulcher (VTF) type”. It is well known that ionic conductivity in a polymer electrolyte is mainly originated from thermal agitation of the polymer chains of the amorphous phase. Temperature dependence of ionic conductivity in an amorphous polymer electrolyte follows a convex curve, and it often shows good correlation with the VTF equation, namely,

$$\sigma = \frac{A}{\sqrt{T}} \exp\left(\frac{-E_a}{T-T_0}\right), \quad (1)$$

where σ is ionic conductivity, A is a constant, E_a (kJ/mol) is apparent activation energy, T (K) is temperature, and T_0 (K) is the temperature at

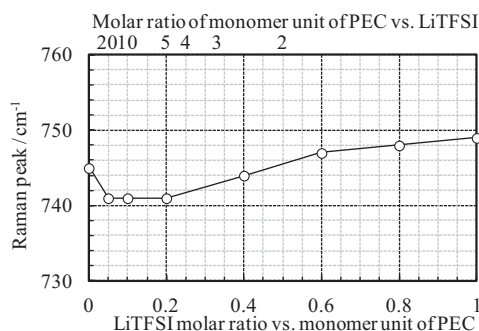


Fig. 8. Salt concentration dependence of the $\delta_s(\text{CF}_3)$ peak for LiTFSI/PEC electrolytes.

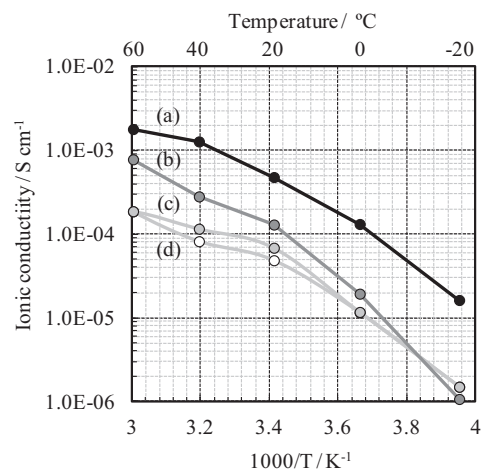


Fig. 9. Arrhenius plots of ionic conductivities for LiTFSI/PEC electrolytes. The values in the figure show the LiTFSI concentration (mol per repeat unit of PEC).

which free volume at apparent temperature T_g (K) is zero. The VTF Eq. (1), agrees well with experimental results across a wide temperature range. It was essentially derived to express the temperature dependence of the viscosity of a super cooled fluid, and linking it with ionic conductivity in a polymer electrolyte shows that ionic conductivity is closely correlated with polymer-segment motion. When the mobility of the polymer chains is high, ionic conductivity is also high. That is, since ionic motion results from the motion of polymer chains, a polymer with low glass-transition temperature, i.e., one with a more flexible framework, will have high ionic conductivity. From that viewpoint, to attain high ionic conductivity, an amorphous low-glass-transition-temperature polymer has been developed. And as for the mechanism of ionic conduction, macromolecular-segment motion contributes to ion transport.

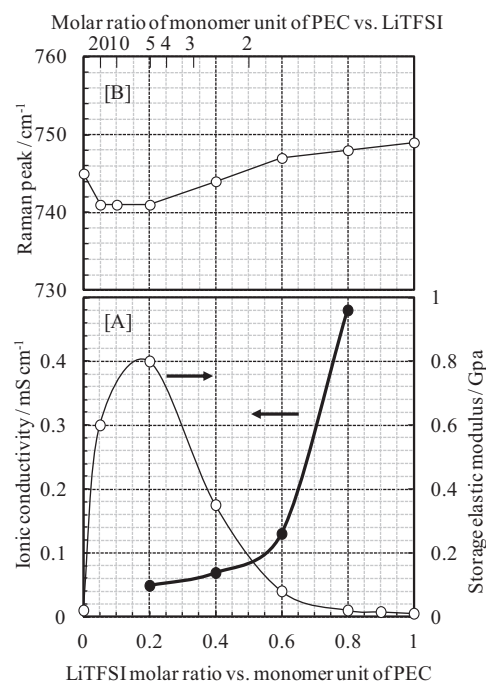


Fig. 10. Salt concentration dependence of (A) ionic conductivity at $20\text{ }^{\circ}\text{C}$ or the storage elastic modulus at $20\text{ }^{\circ}\text{C}$ for LiTFSI/PEC electrolytes, and (B) the $\delta_s(\text{CF}_3)$ peak for LiTFSI/PEC electrolytes.

Fig. 10 shows the LiTFSI concentration dependence of (A) ionic conductivity at 20 °C or the storage elastic modulus at 20 °C for LiTFSI/PEC electrolytes, and (B) the $\delta_s(\text{CF}_3)$ peak for LiTFSI/PEC electrolytes. Up to LiTFSI molar ratio of 0.2, dissociated LiTFSI form macromolecular pseudo-cross-link points, thereby causing storage elastic modulus to increase. Moreover, above LiTFSI mol ratio of 0.2, it is conceivable that polymer-segment movement becomes more active because the abundance ratio of undissociated LiTFSI increases and the salt functions as a plasticizing agent. Therefore, it is considered that the activation of polymer-segmental motion induces high ionic conductivity for PEC based polymer electrolytes. The highest ionic conductivity of 0.47 mS cm^{-1} at 20 °C at LiTFSI mol ratio of 0.8, which is higher than that for PEO based polymer electrolyte, is due to two reasons. One is introduction a carbonate functional group to a polymer framework instead of ether oxygen. As discussed in Table 1, an estimated donor property of a carbonate functional group is lower than that for ether oxygen. We consider that, in the case of a PEC based polymer electrolyte, coupling of a carbonate functional group and a lithium ion is weak, so the polymer electrolyte shows a higher ionic conductivity than that for PEO based polymer electrolyte. The next reason is a high LiTFSI dissolution capability of PEC. The abundance ratio of LiTFSI acts as a plasticizing agent, which induces a high ionic conductivity of a PEC based polymer electrolyte. Though PEC contains a large amount of LiTFSI in the polymer without crystallization of LiTFSI, PEO cannot contain such a large amount of LiTFSI.

The electrochemical stability window of the PEC based polymer electrolyte was evaluated by linear sweep voltammetry using a three-electrode electrochemical cell consisting of a stainless working electrode and a lithium and a counter electrode. Fig. 11 shows the electrochemical stability of the PEC based polymer electrolyte. The lithium-salt concentration in the electrolyte is 0.4 mol per monomer repeating unit of PEC, as a typical example. The onset of current flow at 4.5 V vs. Li/Li^+ is associated with the decomposition of the PEC based polymer electrolyte. This result suggests that the PEC based polymer electrolyte is acceptable for conventional cathode materials such as LiCoO_2 and LiMn_2O_4 .

4. Conclusions

To effectively improve lithium ion conductivity, namely, the product of Li^+ transference number and ion conductivity, this study focused on poly(ethylene carbonate) (PEC) as a base polymer of the polymer electrolytes, instead of conventional poly(ethylene oxide) PEO, which is extensively used as polymer electrolytes, and its key findings are summarized as follows.

Li^+ transference number is higher in the case of the PEC system, namely, 0.4 for PEC versus 0.1 for PEO. In the cases of diethyl ether and diethyl carbonate, the former solvent, containing ^7Li in the ether, exhibits a chemical shift located on the high-magnetic-field side, and ^7Li is more strongly shielded by the ether, so electron density is high. As a result, electrostatic interaction between the ether and lithium is stronger than that between the carbonate and lithium. The solvent with polycarbonate, i.e., PEC, has lower electrostatic interaction with lithium than that of the solvent with polyether, i.e., PEO; consequently, movement of lithium ions is easier, and lithium ion transference number is higher, in the case of PEC.

By combining lithium salt in the PEC up to a concentration of 0.2 mol per repeating unit, it is possible to increase both storage elastic modulus and glass-transition temperature of the polymer electrolyte. This result is considered to be due to formation of pseudo-cross-links by lithium ions dissolving in the polymer and forming coordinate bonds with the polymer. Mixing lithium salt in PEC to concentrations above 0.2 mol

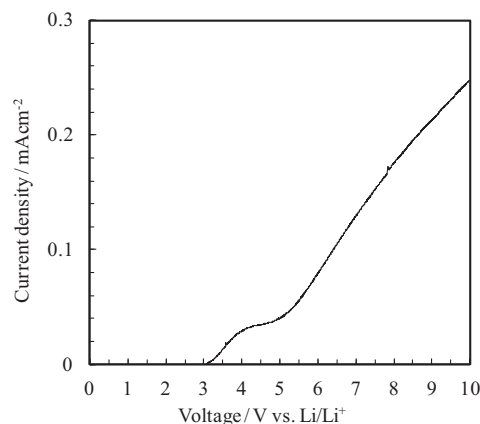


Fig. 11. Linear sweep voltammetry for LiTFSI/PEC electrolytes. LiTFSI concentration per repeating unit of PEC is 0.4.

per repeating unit decreases both storage elastic modulus and glass-transition temperature of the polymer electrolyte. In that concentration range, the abundance ratio of undissociated salt is high, and growth of the pseudo-cross-links is suppressed. Moreover, since storage elastic modulus is decreased, the salt is thought to act as a plasticizing agent.

By mixing lithium salt in PEC to concentration of 0.8 mol per repeating unit, it was possible to attain an ion conductivity of 0.47 mS cm^{-1} .

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